Take $u = (\log 2)x$ so that

$$y = e^u$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^u \times \log 2 = \log 2\, e^{x \log 2}$$

$$= (\log 2) 2^x$$

By using the differentiation formula for a^x, one can find the derivative directly.

Example 10.16

If $y = \tan^{-1}\left(\frac{1+x}{1-x}\right)$, find y'.

Solution

$$y = \tan^{-1}\left(\frac{1+x}{1-x}\right)$$

Let $\frac{1+x}{1-x} = t$

Then, $y = \tan^{-1} t$

$$\frac{dy}{dx} = \frac{d}{dt}(\tan^{-1} t).\frac{dt}{dx}$$

$$= \frac{1}{1+t^2}.\frac{(1-x).1-(1+x)(-1)}{(1-x)^2}$$

$$= \frac{1}{1+\left(\frac{1+x}{1-x}\right)^2}.\frac{(1-x)+(1+x)}{(1-x)^2} = \frac{1}{1+x^2}.$$

EXERCISE 10.3

Differentiate the following :

(1) $y = (x^2+4x+6)^5$ (2) $y = \tan 3x$ (3) $y = \cos(\tan x)$

(4) $y = \sqrt[3]{1+x^3}$ (5) $y = e^{\sqrt{x}}$ (6) $y = \sin(e^x)$

(7) $F(x) = (x^3+4x)^7$ (8) $h(t) = \left(t - \frac{1}{t}\right)^{\frac{3}{2}}$ (9) $f(t) = \sqrt[3]{1+\tan t}$

(10) $y = \cos(a^3 + x^3)$ (11) $y = e^{-mx}$ (12) $y = 4 \sec 5x$

(13) $y = (2x-5)^4(8x^2-5)^{-3}$ (14) $y = (x^2+1)\sqrt[3]{x^2+2}$ (15) $y = xe^{-x^2}$

(16) $s(t) = \sqrt[4]{\frac{t^3+1}{t^3-1}}$ (17) $f(x) = \frac{x}{\sqrt{7-3x}}$ (18) $y = \tan(\cos x)$

(19) $y = \frac{\sin^2 x}{\cos x}$ (20) $y = 5^{\frac{-1}{x}}$ (21) $y = \sqrt{1+2\tan x}$

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